HAT-P-32b

R-0052

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_171023` · created 2026-07-09T07:30:46+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458049.6614 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.003 +/- 0.043
Transit depth (Rp/Rs)^2	0.00067 +/- 0.022 [%]
Orbital Inclination (inc)	88.79 +/- 1.72
Airmass coefficient 1 (a1)	1.28 +/- 0.12
Airmass coefficient 2 (a2)	-0.182 +/- 0.031
Scatter in the residuals of the lightcurve fit is	9.21 %
Best Comparison Star	None
Optimal Aperture	5.01
Optimal Annulus	5.01
Transit Duration (day)	0.1155 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.003 ± 0.043 vs 0.1489 (deviation 3.39σ)
Duration fit vs published	0.1155 d vs 0.1304 d (deviation 4.39σ)
Mid-time O–C	0.8 min
Depth SNR	0.1
Residual scatter	9.21 %

Light curve

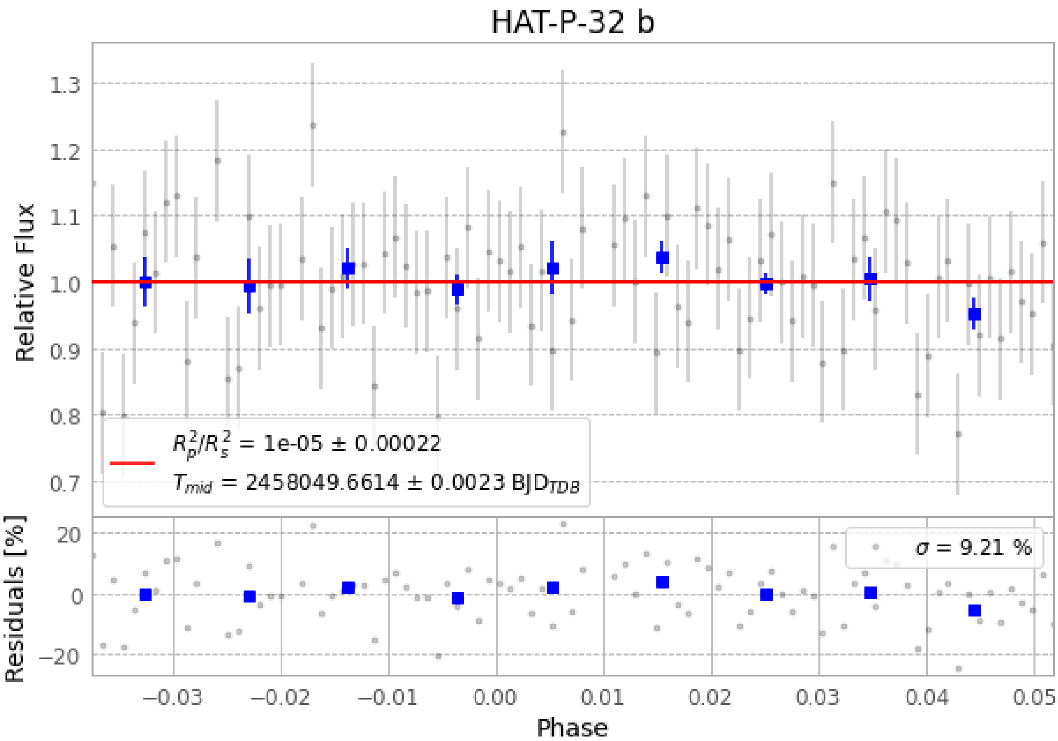

Figure 1 — FinalLightCurve_HAT-P-32 b_2017-10-22.png (SHA-256 f43ec44139edb472…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 241], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a07a50557eb5d91e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

91 FITS frames (60 MB), `HATP-32171023014812.FITS` → `HATP-32171023062412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-171023.log (24326bb5ab45…), FinalLightCurve_HAT-P-32 b_2017-10-22.png (f43ec44139ed…), FinalLightCurve_HAT-P-32 b_2017-10-22.csv (795a5eccc526…)

Compute resources

Wall time 147.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-15 16:40Z.